57. SRAM

57.1 Overview

The MCU provides an on-chip, high-density SRAM module with Error Correction Code (ECC).

In single-core, C-TCM and S-TCM at addresses 0x2A00_0000 to 0x2A01_FFFF can be used as SRAM.

See the section 2, CPU for details on TCM.

Table 57.1 SRAM specifications

Parameter		SRAM0	SRAM1	SRAM2	SRAM3
SRAM capacity		512 KB	512 KB	512 KB	128 KB
SRAM address	Secure alias	0x2200_0000 to 0x2207_FFFF	0x2208_0000 to 0x220F_FFFF	0x2210_0000 to 0x2217_FFFF	0x2218_0000 to 0x2219_FFFF
	Non-secure alias	0x3200_0000 to 0x3207_FFFF	0x3208_0000 to 0x320F_FFFF	0x3210_0000 to 0x3217_FFFF	0x3218_0000 to 0x3219_FFFF
ECC region[*1]	Secure alias	0x221A_0000 to 0x221A_FFFF	0x221B_0000 to 0x221B_FFFF	0x221C_0000 to 0x221C_FFFF	0x221D_0000 to 0x221D_3FFF
	Non-secure alias	0x321A_0000 to 0x321A_FFFF	0x321B_0000 to 0x321B_FFFF	0x321C_0000 to 0x321C_FFFF	0x321D_0000 to 0x321D_3FFF
Access		Wait states are not inserted into the access cycle by default. If the ICLK frequency is greater than half the maximum frequency in the electrical characteristics, a wait state is required. If the ICLK frequency is less than or equal half the maximum frequency in the electrical characteristics, a wait state is not required.			
Data retention function		Not available in Deep Software Standby mode			
Module-stop function		Module-stop state can be set to reduce power consumption			
Error checking		SEC-DED (Single-Error Correction and Double-Error Detection Code: 64-bit data with 8-bit ECC code)			
Security		TrustZone Filter is integrated for memory access and SFR access. Access to the memory space is controlled by setting the memory Security Attribution (SA). And access to I/O space (SFR) space is controlled by setting the register Security Attribution (SA). See section 57.3.5. TrustZone Filter Function.			

Note 1. When ECC function is disabled, it is used as a data region and can be accessed directly. When ECC function is enabled, direct access is disabled. When bypass is enabled, direct access to ECC bit is possible.

57.2 Register Descriptions

57.2.1 SRAMSABARn : SRAM Security Attribute Boundary Address Register (n = 0 to 3)

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x400 + 0x04 × n

Bit position: 31 … 0

Bit field: (no named fields)

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Boundary address between secure and non-secure. (Offset of the start address of non-secure region)	R/W

Note: S-TYPE-1, P-TYPE-1
Note: This register is write-protected by the PRCR_S.PRC4 bit.

SRAMSABARn specify the offset of the boundary between the secure and non-secure regions of each SRAM. Write the absolute address to the SRAMSABARn. When writing to SRAMSABARn, writing from b31 to b21 is ignored and the value written from b12 to b0 should be 0.

The region lower than the boundary address is marked as "Secure", and higher than or equal to the boundary address is marked as "Non-secure".

The boundary address is as follows:

0x2200_0000 + SRAMSABARn (Secure alias)

0x3200_0000 + SRAMSABARn (Non-secure alias)

For SRAMSABAR0

When the offset of the boundary address is 0x0000_0000, whole SRAM0 is marked as Non-secure.

When the offset of the boundary address is 0x0008_0000 or large, whole SRAM0 is marked as Secure.

For SRAMSABAR1

When the offset of the boundary address is 0x0008_0000 or less, whole SRAM1 is marked as Non-secure.

When the offset of the boundary address is 0x0010_0000 or large, whole SRAM1 is marked as Secure.

For SRAMSABAR2

When the offset of the boundary address is 0x0010_0000 or less, whole SRAM2 is marked as Non-secure.

When the offset of the boundary address is 0x0018_0000 or large, whole SRAM2 is marked as Secure.

For SRAMSABAR3

When the offset of the boundary address is 0x0018_0000 or less, whole SRAM3 is marked as Non-secure.

When the offset of the boundary address is 0x001A_0000 or large, whole SRAM3 is marked as Secure.

57.2.2 SRAMSAR : SRAM Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SRAMWTSA	—	—	—	—	SRAMSA3	SRAMSA2	SRAMSA1	SRAMSA0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SRAMSA0	SRAM0 Register Security Attribution 0: Secure 1: Non-Secure	R/W
1	SRAMSA1	SRAM1 Register Security Attribution 0: Secure 1: Non-Secure	R/W
2	SRAMSA2	SRAM2 Register Security Attribution 0: Secure 1: Non-Secure	R/W
3	SRAMSA3	SRAM3 Register Security Attribution 0: Secure 1: Non-Secure	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	SRAMWTSA	SRAMWTSC Security Attribution 0: Secure 1: Non-Secure	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1
Note: Only Secure access can write to this register. Both Secure access and Non-secure read access are allowed but Non-secure write access is not allowed and TrustZone access error is not generated.
Note: This register is write-protected by the PRCR_S.PRC4 bit.

SRAMSA0 bit (SRAM0 Register Security Attribution)

Security attributes of SRAMCR0 registers. The target registers are as follows.

- SRAMCR0
- SRAMECCRGN0
- SRAMESCLR.CLR00, SRAMESCLR.CLR01

SRAMSA1 bit (SRAM1 Register Security Attribution)

Security attributes of SRAMCR1 registers. The target registers are as follows.

- SRAMCR1
- SRAMECCRGN1
- SRAMESCLR.CLR10, SRAMESCLR.CLR11

SRAMSA2 bit (SRAM2 Register Security Attribution)

Security attributes of SRAMCR2 registers. The target registers are as follows.

- SRAMCR2
- SRAMECCRGN2
- SRAMESCLR.CLR20, SRAMESCLR.CLR21

SRAMSA3 bit (SRAM3 Register Security Attribution)

Security attributes of SRAMCR3 registers. The target registers are as follows.

- SRAMCR3
- SRAMECCRGN3
- SRAMESCLR.CLR30, SRAMESCLR.CLR31

SRAMWTSA bit (SRAMWTSC Security Attribution)

Security attributes of SRAMWTSC.

57.2.3 SRAMESAR : SRAM ECC region Security Attribute Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x510

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SRAM ESA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SRAMESA	ECC region Security Attribution 0: Secure 1: Non-Secure	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1
Note: This register is write-protected by PRCR_S.PRC4.

SRAMESA bit (ECC region Security Attribution)

This bit indicates the security attribute of the ECC region of all SRAMs.

57.2.4 SRAMPRCR_S : SRAM Protection Control Register for Secure

Base address: SRAM = 0x4000_2000

Offset address: 0x00

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KW[7:0]								—	—	—	—	—	—	—	PR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PR	Register Write Control 0: Writing to registers are disabled 1: Writing to registers are enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KW[7:0]	Write Key Code Key code protection to the PR bit	W

Note: S-TYPE-6, P-TYPE-2
Note: It is necessary to write by half word access.
Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

PR bit (Register Write Control)

The PR bit controls the write mode of the SRAMWTSC, SRAMCRn, SRAMECCRGNn registers that are marked as “Secure” by SRAMSAR register. When this bit is set to 1, writing to the SRAMWTSC, SRAMCRn, SRAMECCRGNn marked as “Secure” is enabled.

While writing to this register it is necessary to write 0xA5 to the KW[7:0] bits simultaneously.

KW[7:0] bits (Write Key Code)

The KW[7:0] bits enable or disable writing to the PR bit. When you write to the PR bit, write 0xA5 to the KW[7:0] bits simultaneously. When a value other than 0xA5 is written to KW[7:0], the PR bit is not updated. The KW[7:0] bits are always read as 0x00.

57.2.5 SRAMPRCR_NS : SRAM Protection Control Register for Non-Secure

Base address: SRAM_NS = 0x5000_2000

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KW[7:0]								—	—	—	—	—	—	—	PR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PR	Register Write Control 0: Writing to registers are disabled 1: Writing to registers are enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KW[7:0]	Write Key Code Key code protection to the PR bit	W

Note: S-TYPE-7, P-TYPE-2
Note: It is necessary to write by half word access.
Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

PR bit (Register Write Control)

The PR bit controls the write mode of the SRAMWTSC, SRAMCRn, SRAMECCRGNn registers that are marked as "Non-secure" by SRAMSAR register. When this bit is set to 1, writing to the SRAMWTSC, SRAMCRn, SRAMECCRGNn marked as "Non-secure" is enabled.

While writing to this register it is necessary to write 0xA5 to the KW[7:0] bits simultaneously.

KW[7:0] bits (Write Key Code)

The KW[7:0] bits enable or disable writing to the PR bit. When you write to the PR bit, write 0xA5 to the KW[7:0] bits simultaneously. When a value other than 0xA5 is written to KW[7:0], the PR bit is not updated. The KW[7:0] bits are always read as 0x00.

57.2.6 SRAMWTSC : SRAM Wait State Control Register

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	WTEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	WTEN	SRAM wait enable 0: Do not add wait state in the access cycle to SRAMs. 1: Add wait state in the access cycle to SRAMs.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3 (SRAMSAR.SRAMWTSA), P-TYPE-2
Note: SRAMWTSC is valid only for SRAM0, 1, 2, 3.
Note: The protection register (SRAMPRCR_S or SRAMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the SRAMPRCR_S or SRAMPRCR_NS register is 1.

WTEN bit (SRAM wait enable)

This bit sets the wait cycle insertion to the access cycle of SRAM. When it is set 1 to the WTEN bit, 1 wait cycle inserted to the read access cycle of SRAM. Also When the WTEN bit is 1, and access to the same region*1 and the same mat*2 is continuous, 1 wait cycle is inserted to the second access cycle. When ICLK frequency is greater than half the maximum frequency in the electrical characteristics, it is necessary to set 1 wait cycle in the WTEN bit.

Note 1. The region of SRAM is divided to 128KB units. The ECC region is not divided.
Note 2. The mat is divided to 0x0 to 0x7 and 0x8 to 0xF in the lower 4-bits of the address.

For example, in the case of SRAMn, the same region and the same mat is 8 areas of SRAM region, and 2 areas of ECC region in the figure below. If access 1 area continuously, 1 wait cycle is inserted. If ECC function is enabled or bypass is enabled, ECC region is accessed, so if access to the same mat continues, 1 wait cycle is inserted.

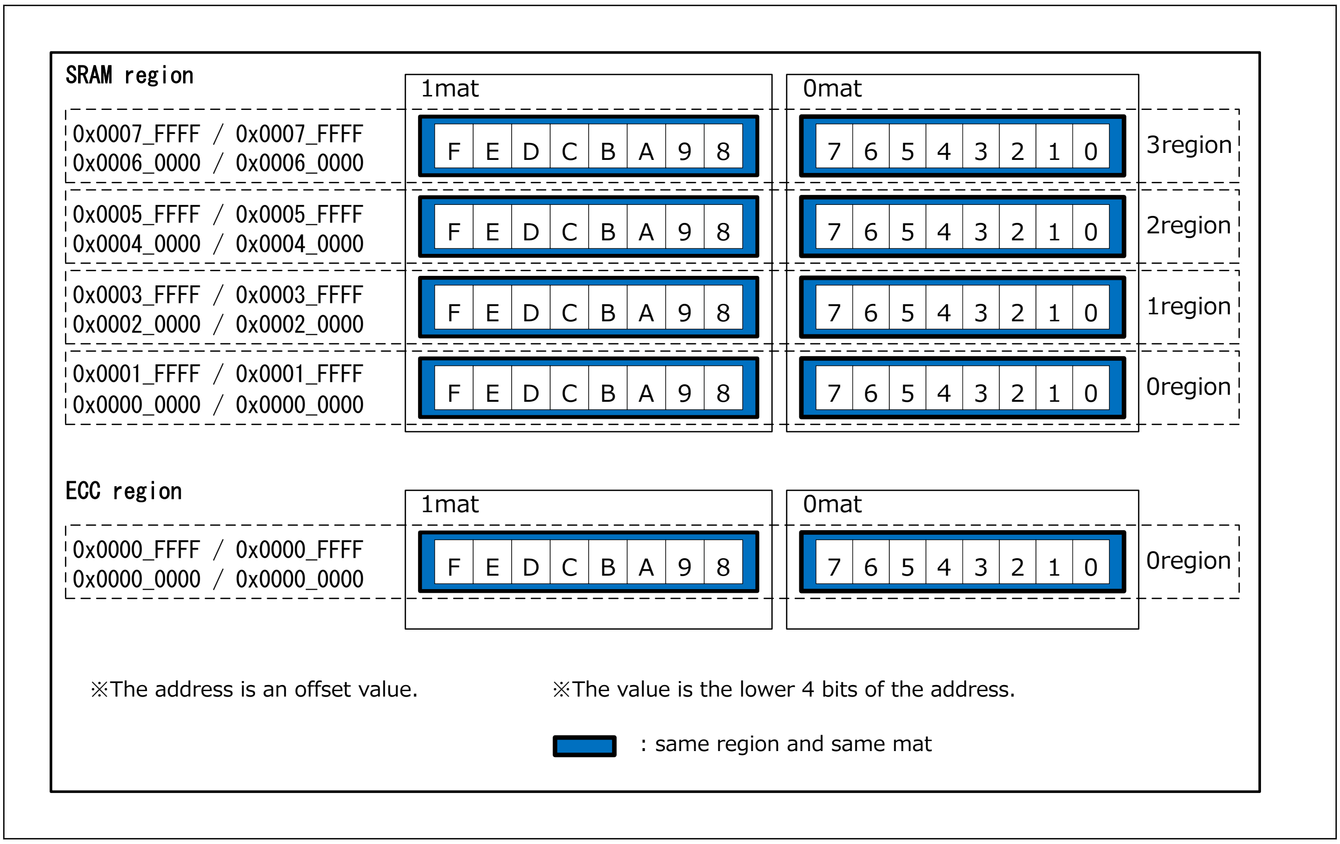

Figure 57.1 Mat configuration of each SRAMs

57.2.7 SRAMCRn : SRAM Control Register n For ECC RAM (n = 0 to 3)

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x10 + 0x04 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TSTBYP	—	—	E1STSEN	ECCMOD[1:0]		—	OAD
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OAD	Operation after error detection 0: Interrupt 1: Reset	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
3:2	ECCMOD[1:0]	ECC Operating Mode Select 0 0: Disable ECC function 0 1: Setting prohibited 1 0: Enable ECC function without error checking 1 1: Enable ECC function with error checking	R/W
4	E1STSEN	ECC 1-bit Error Update Enable 0: Disable updating of 1-bit ECC error 1: Enable updating of 1-bit ECC error	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	TSTBYP	ECC Test Enable / ECC Bypass Select 0: Disable ECC bypass 1: Enable ECC bypass	R/W

Note: S-TYPE-3 (SRAMSAR.SRAMSA0), P-TYPE-2

Note: The protection register (SRAMPRCR_S or SRAMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the SRAMPRCR_S or SRAMPRCR_NS register is 1.

If CPU0 and CPU1 issue speculative access to an uninitialized SRAM with ECC enabled, it may trigger an interrupt or reset. After enabling ECC function with error checking, initialize the SRAM before setting the interrupt in the IELSRn register or setting the reset in the OAD bit.

OAD bit (Operation after error detection)

This bit selects a reset or non-maskable interrupt when an ECC error is detected.

ECCMOD[1:0] bits (ECC Operating Mode Select)

These bits set the access mode to the ECC operating.

E1STSEN bit (ECC 1-bit Error Update Enable)

This bit enables or disables updating of the ECC 1-bit error status bit in the SRAMESR, in response to a 1-bit error. This bit also functions as an interrupt or a reset mask.

TSTBYP bit (ECC Test Enable / ECC Bypass Select)

This bit enables direct access to ECC code by bypassing the ECC function. The ECC Bypass function is used along with setting the ECCMOD[1:0] bits in the same register to 00b.

For “64-bit data with 8-bit ECC code”

The ECC must be accessed the same address. The ECC code is assigned to the lower 8 bits. When writing the ECC code, the bits above the ECC bits are ignored. When reading the ECC code, the bits above the ECC bits are invalid.

Note: For details about ECC test, see section 57.3.4. ECC Decoder Testing.

57.2.8 SRAMECCRGN0 : SRAM ECC Region Control Register 0

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x30

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ECCRGN[2:0]		
Value after reset:	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
2:0	ECCRGN[2:0]	ECC target region select The ECC target region selected by this register can be controlled by SRAMCR0. 0 0 0: No ECC target region 0 0 1: 0x2200_0000–0x2201_FFFF 0x3200_0000–0x3201_FFFF (128 KB) 0 1 0: 0x2200_0000–0x2203_FFFF 0x3200_0000–0x3203_FFFF (256 KB) 0 1 1: 0x2200_0000–0x2205_FFFF 0x3200_0000–0x3205_FFFF (384 KB) 1 0 0: 0x2200_0000–0x2207_FFFF 0x3200_0000–0x3207_FFFF (512 KB) Others: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3 (SRAMSAR.SRAMSAn), P-TYPE-2
Note: The protection register (SRAMPRCR_S or SRAMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the SRAMPRCR_S or SRAMPRCR_NS register is 1.

ECCRGN[2:0] bits (ECC target region select)

These bits select ECC target region in SRAM0.

57.2.9 SRAMECCRGN1 : SRAM ECC Region Control Register 1

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x34

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ECCRGN[2:0]		
Value after reset:	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
2:0	ECCRGN[2:0]	ECC target region select The ECC target region selected by this register can be controlled by SRAMCR1. 0 0 0: No ECC target region 0 0 1: 0x2208_0000–0x2209_FFFF 0x3208_0000–0x3209_FFFF (128 KB) 0 1 0: 0x2208_0000–0x220B_FFFF 0x3208_0000–0x320B_FFFF (256 KB) 0 1 1: 0x2208_0000–0x220D_FFFF 0x3208_0000–0x320D_FFFF (384 KB) 1 0 0: 0x2208_0000–0x220F_FFFF 0x3208_0000–0x320F_FFFF (512 KB) Others: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3 (SRAMSAR.SRAMSAn), P-TYPE-2
Note: The protection register (SRAMPRCR_S or SRAMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the SRAMPRCR_S or SRAMPRCR_NS register is 1.

ECCRGN[2:0] bits (ECC target region select)

These bits select ECC target region in SRAM1.

57.2.10 SRAMECCRGN2 : SRAM ECC Region Control Register 2

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x38

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ECCRGN[2:0]		
Value after reset:	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
2:0	ECCRGN[2:0]	ECC target region select The ECC target region selected by this register can be controlled by SRAMCR2. 0 0 0: No ECC target region 0 0 1: 0x2210_0000–0x2211_FFFF 0x3210_0000–0x3211_FFFF (128 KB) 0 1 0: 0x2210_0000–0x2213_FFFF 0x3210_0000–0x3213_FFFF (256 KB) 0 1 1: 0x2210_0000–0x2215_FFFF 0x3210_0000–0x3215_FFFF (384 KB) 1 0 0: 0x2210_0000–0x2217_FFFF 0x3210_0000–0x3217_FFFF (512 KB) Others: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3 (SRAMSAR.SRAMSAn), P-TYPE-2
Note: The protection register (SRAMPRCR_S or SRAMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the SRAMPRCR_S or SRAMPRCR_NS register is 1.

ECCRGN[2:0] bits (ECC target region select)

These bits select ECC target region in SRAM2.

57.2.11 SRAMECCRGN3 : SRAM ECC Region Control Register 3

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x3C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ECCRGN[2:0]		
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
2:0	ECCRGN[2:0]	ECC target region select The ECC target region selected by this register can be controlled by SRAMCR3. 0 0 0: No ECC target region 0 0 1: 0x2218_0000–0x2219_FFFF 0x3218_0000–0x3219_FFFF (128 KB) Others: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3 (SRAMSAR.SRAMSAn), P-TYPE-2

Note: The protection register (SRAMPRCR_S or SRAMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the SRAMPRCR_S or SRAMPRCR_NS register is 1.

ECCRGN[2:0] bits (ECC target region select)

These bits select ECC target region in SRAM3.

57.2.12 SRAMESR : SRAM Error Status Register For ECC RAM

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x40

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	ERR31	ERR30	ERR21	ERR20	ERR11	ERR10	ERR01	ERR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ERR00	SRAM0 1-bit ECC Error Status 0: 1-bit ECC error has not occurred. 1: 1-bit ECC error has occurred.	R
1	ERR01	SRAM0 2-bit ECC Error Status 0: 2-bit ECC error has not occurred. 1: 2-bit ECC error has occurred.	R
2	ERR10	SRAM1 1-bit ECC Error Status 0: 1-bit ECC error has not occurred. 1: 1-bit ECC error has occurred.	R
3	ERR11	SRAM1 2-bit ECC Error Status 0: 2-bit ECC error has not occurred. 1: 2-bit ECC error has occurred.	R
4	ERR20	SRAM2 1-bit ECC Error Status 0: 1-bit ECC error has not occurred. 1: 1-bit ECC error has occurred.	R

Bit	Symbol	Function	R/W
5	ERR21	SRAM2 2-bit ECC Error Status 0: 2-bit ECC error has not occurred. 1: 2-bit ECC error has occurred.	R
6	ERR30	SRAM3 1-bit ECC Error Status 0: 1-bit ECC error has not occurred. 1: 1-bit ECC error has occurred.	R
7	ERR31	SRAM3 2-bit ECC Error Status 0: 2-bit ECC error has not occurred. 1: 2-bit ECC error has occurred.	R
15:8	—	These bits are read as 0.	R

Note: S-TYPE-5, P-TYPE-2
Note: This register is initialized after returning from software standby mode while the debugger is connected.

This register is cleared by the corresponding bit in SRAMESCLR register or resets other than Bus Error Reset and Memory Error Reset. If error and clear occur at the same time, clear has priority. Also during accessing from debugger it is to stop updating register.

ERRn0 bit (SRAMn 1-bit ECC Error Status)

This bit shows whether there is a 1-bit ECC error in SRAMn.

When ECC function with error checking is enabled and, if the SRAMCRn.E1STSEN bit is 1, the ERRn0 bit is set to 1 if a 1-bit error is detected.

When ERRn0 bit is set to 1, outputs a reset or an interrupt request according to the SRAMCRn.OAD bit.

ERRn1 bit (SRAMn 2-bit ECC Error Status)

This bit shows whether there is a 2-bit ECC error in SRAMn.

When ECC function with error checking is enabled, the ERRn1 bit is set to 1 if a 2-bit error is detected.

When ERRn1 bit is set to 1, outputs a reset or an interrupt request according to the SRAMCRn.OAD bit.

57.2.13 SRAMESCLR : SRAM Error Status Clear Register For ECC RAM

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x48

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CLR31	CLR30	CLR21	CLR20	CLR11	CLR10	CLR01	CLR00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLR00	SRAM0 1-bit ECC Error Status Clear Writing to the CLR00 clears SRAMESR.ERR00. 0: No effect 1: Clear 1-bit ECC error	W
1	CLR01	SRAM0 2-bit ECC Error Status Clear Writing to the CLR01 clears SRAMESR.ERR01. 0: No effect 1: Clear 2-bit ECC error.	W
2	CLR10	SRAM1 1-bit ECC Error Status Clear Writing to the CLR10 clears SRAMESR.ERR10. 0: No effect 1: Clear 1-bit ECC error.	W

Bit	Symbol	Function	R/W
3	CLR11	SRAM1 2-bit ECC Error Status Clear Writing to the CLR11 clears SRAMESR.ERR11. 0: No effect 1: Clear 2-bit ECC error.	W
4	CLR20	SRAM2 1-bit ECC Error Status Clear Writing to the CLR20 clears SRAMESR.ERR20. 0: No effect 1: Clear 1-bit ECC error.	W
5	CLR21	SRAM2 2-bit ECC Error Status Clear Writing to the CLR21 clears SRAMESR.ERR21. 0: No effect 1: Clear 2-bit ECC error.	W
6	CLR30	SRAM3 1-bit ECC Error Status Clear Writing to the CLR30 clears SRAMESR.ERR30. 0: No effect 1: Clear 1-bit ECC error.	W
7	CLR31	SRAM3 2-bit ECC Error Status Clear Writing to the CLR31 clears SRAMESR.ERR31. 0: No effect 1: Clear 2-bit ECC error.	W
15:8	—	The write value should be 0.	W

Note: S-TYPE-4 (SRAMSAR.SRAMSAn), P-TYPE-2

CLRn0 bit (SRAMn 1-bit ECC Error Status Clear)

This bit can clear 1-bit ECC error status bit in the SRAMESR. If error and clear occur at the same time, clear has priority.

SRAMESR.ERRn0 can be cleared, when writing 1 to CLRn0.

CLRn1 bit (SRAMn 2-bit ECC Error Status Clear)

This bit can clear 2-bit ECC error status bit in the SRAMESR. If error and clear occur at the same time, clear has priority.

SRAMESR.ERRn1 can be cleared, when writing 1 to CLRn1.

57.2.14 SRAMEARnm : SRAM Error Address Register nm For ECC RAM (n = 0 to 3, m = 0 to 1)

Base address: SRAM = 0x4000_2000
SRAM_NS = 0x5000_2000

Offset address: 0x50 + 0x10 × n + 0x04 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	n/a	When an SRAM error occurs, it stores offset of an error address.	R

Note: S-TYPE-5, P-TYPE-2
Note: This register is initialized after returning from software standby mode while the debugger is connected.

This register is cleared by the corresponding bit in SRAMESCLR register, or resets other than Bus Error Reset and Memory Error Reset. If error and clear occur at the same time, clear has priority. Also during accessing from debugger it is to stop updating register.

The error address is as follows.

0x2200_0000 + SRAMEARnm (Secure alias)

0x3200_0000 + SRAMEARnm (Non-secure alias)

For SRAMEARn0

This register stores offset of the error address where 1-bit ECC error is detected. These bits hold offset of the error address that occurred first. These bits are cleared by clearing 1-bit ECC error from SRAMESCLR.

For SRAMEARn1

This register stores offset of the error address where 2-bit ECC error is detected. These bits hold offset of the error address that occurred first. These bits are cleared by clearing 2-bit ECC error from SRAMESCLR.

57.3 Operation

57.3.1 Module Stop Function

Power consumption can be reduced by setting module stop control register A (MSTPCRA) to stop supply of the clock signal to SRAM.

SRAMn (n = 0, 1, 2, 3) is controlled by SRAMn bit in MSTPCRA register and, in the case of 1, SRAMn becomes the clock stop state.

The SRAM is thus placed in the module-stop state by stopping supply of the clock signals. The SRAM operates after a reset.

SRAM is not accessible if it is in the module-stop state. A transition to the module-stop state should not be made while access to SRAM is in progress.

Access to the SRAM in the module-stop state is prohibited. If access is attempted, correct operation is not guaranteed.

For details on the MSTPCRA register, see section 11, Low Power Mode.

57.3.2 Correction of ECC Errors

Enabling and disabling of ECC error correction can be selected through the setting by ECCMOD[1:0] bits of SRAMCRn register. In the initial state, ECC error correction is disabled. The ECC check type is SEC-DED (Single-Error Correction and Double-Error Detection Code).

When ECC function is enabled, 8-bit check bits are appended to 64-bit data for writing. For reading, 72-bit (data: 64 bits, check bits: 8 bits) data is read out from the SRAM (ECC area).

When ECC function with error checking are enabled, error correction is done if a 1-bit error occurs and the ERRn0 bit in the SRAMESR register is set to 1 if the E1STSEN bit in the SRAMCRn register is 1. If a 2-bit error occurs, error detection is done and the ERRn1 bit in the SRAMESR register is set to 1, though error correction is not performed.

When ECC function without error checking is enable, error correction is done if a 1-bit error occurs but ERRn0 bit in the SRAMESR register is not updated although E1STSEN bit in the SRAMCRn register is 1. If a 2-bit error occurs, this error is detected but the ERRn1 bit in the SRAMESR register is not updated, and error correction is not performed.

When ECC function is disable, neither error correction nor error detection is done although 1-bit or 2-bit error occur.

So ERRn0 and ERRn1 bits are not updated.

When updating all the data after the occurrence of an error, the only support of 64 bit data writing.

Since the SRAM data is undefined after power on and release from Deep Software Standby mode, accessing the SRAM when ECC function is enabled and error checking is selected causes an ECC error to occur. Therefore, before using ECC function, initial writing with ECC function enabled and 64-bit data size, or initial writing with ECC function enabled and the error checking disabled, to the area to be used in the SRAM should be done.

57.3.3 ECC Error Interrupt Function

When ECC function with error checking is enabled, error checking is possible in the SRAMn. In the case of a 2-bit error, the ERRn1 bit in the SRAMESR register set to 1. In the case of a 1-bit error, the ERRn0 bit in the SRAMESR register set to 1 when SRAMCRn.E1STSEN bit is 1.

When the ECC 1-bit error is to be masked, it is necessary to set the SRAMCRn.E1STSEN bit to 0 to disable updating to 1 of the SRAMESR.ERRn0 bit. An ECC error will not be generated while ECC function is disabled or ECC function without error checking is enabled.

ECC error may result in interrupt or reset. When ECC error occurs, When the SRAMCRn.OAD bit is 1, outputs a reset request. When it is 0, outputs a non-maskable interrupt request to the ICU.

ECC Error interrupt occurs when one flag in the SRAMESR register is set. ECC Error interrupt continues to occur until the SRAMESR register flag is cleared.

57.3.4 ECC Decoder Testing

Figure 57.2 shows the ECC decoder testing flowchart.

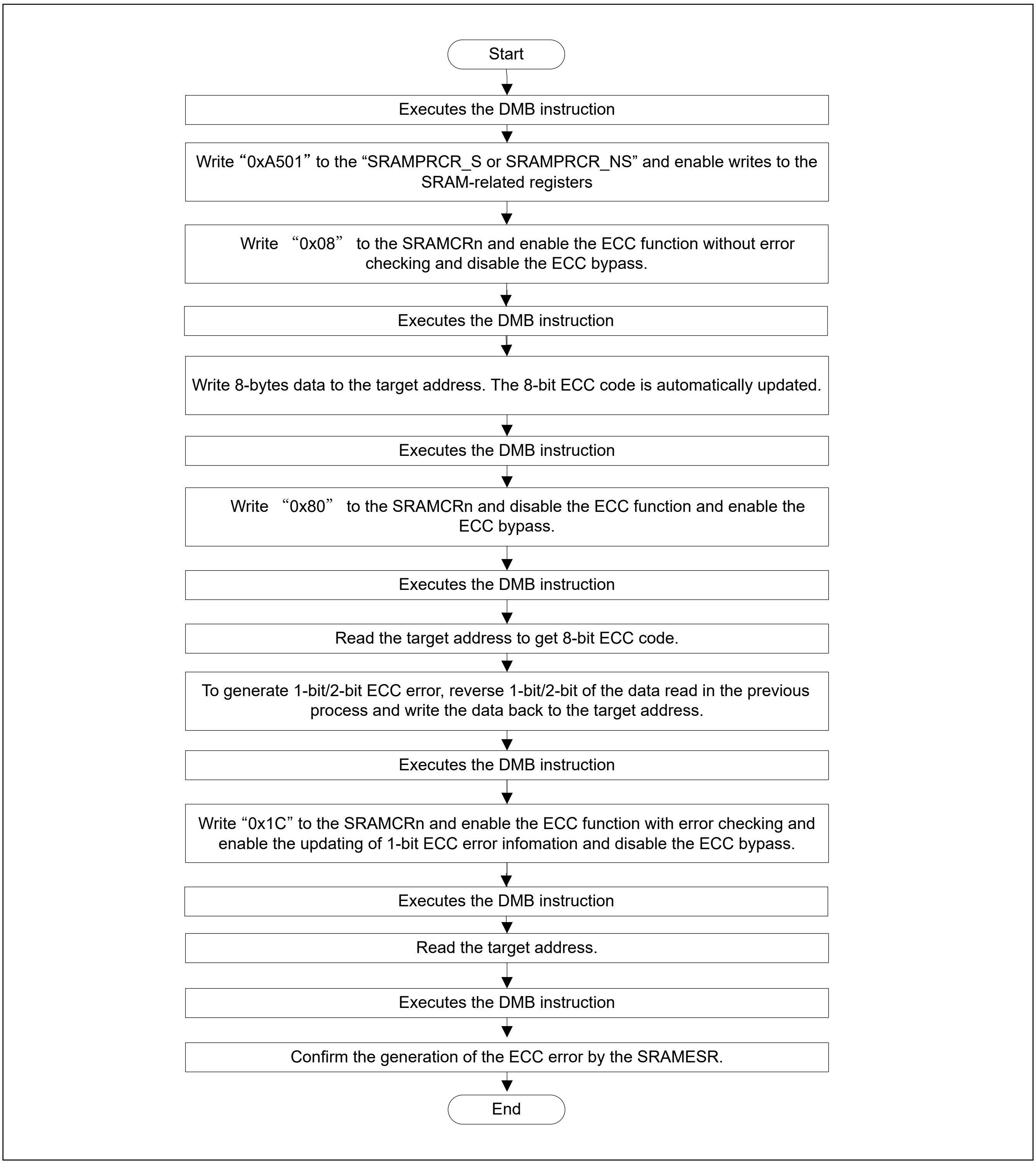

Figure 57.2 ECC decoder testing

57.3.5 TrustZone Filter Function

57.3.5.1 TrustZone Filter for SRAM Registers

SRAM registers can be protected with a Security Attribution (SA) from Non-secure or secure transaction. When SA indicates that SRAM registers are secure status, non-secure transaction can not access them because TrustZone Filter detects an error and protects the access. When SA indicates that SRAM registers are non-secure status, secure transaction can not access them because Trust Zone Filter detects an error and protects the access. SA for SRAM registers is just one to be used commonly among SRAM registers.

In the case of a failed access due to TrustZone error, error response is generated.

Table 57.2 Register protection

SA	Transaction	Write access	Read access
Secure	Secure	Permitted	Permitted
	Non-secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
Non-secure	Secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
	Non-secure	Permitted	Permitted

57.3.5.2 TrustZone Filter for SRAM Memory Regions

SRAM0, SRAM1, SRAM2, SRAM3 regions can divide into secure, non-secure independently. The access permissions for these regions are as below.

Table 57.3 Memory protection

SA	Transaction	Write access	Read access
Secure	Secure	Permitted	Permit
	Non-secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
Non-secure	Secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
	Non-secure	Permitted	Permitted

TrustZone Filter error for SRAM memory, generates an error notification.

57.3.6 Interrupt Source

The SRAM interrupt source includes an ECC error and TrustZone filter error. ECC error can choose interrupt or reset by the SRAMCRn.OAD bit. The SRAM interrupt occurs when one error status in the SRAMESR register is set to 1, the SRAM interrupt continues to occur until the SRAMESR register flag is cleared. When Common memory error occur (NMISR.CMST = 1 or RSTSR1.CMRF = 1), read SRAMESR and check SRAM interrupt source. When access is from the debugger, the error is detected and corrected, but no error flag is set, and no reset request and interrupt request is output.

Table 57.4 SRAM Interrupt Source

Name	Interrupt Source	DTC Activation	DMAC Activation
ECCERR	ECC error (SRAMs with ECC)	Not possible	Not possible
TZFLT	TrustZone filter error	Not possible	Not possible

57.3.7 Wait State

When ICLK frequency is greater than half the maximum frequency in the electrical characteristics do not set 0x00 in wait enable bit for the SRAMWTSC register, in order to insert a wait cycle.

When the wait is not inserted, the operation is not guaranteed.

When ICLK frequency is less than or equal half the maximum frequency in the electrical characteristics wait state is not required.

When the maximum ICLK frequency is 250MHz

- 250MHz ≥ ICLK > 125MHz = 1 wait
- 125MHz ≥ ICLK = No-wait

When the maximum ICLK frequency is 200MHz

- 200MHz ≥ ICLK > 100MHz = 1 wait
- 100MHz ≥ ICLK = No-wait

When the maximum ICLK frequency is 150MHz

- 150MHz ≥ ICLK > 75MHz = 1 wait
- 75MHz ≥ ICLK = No-wait

57.3.8 Access Cycle

The following shows the delay information of the access cycle.

- When ECC is enabled, writing less than 64 bits is 2 cycles slower than normal writing.
- Insert the wait cycle according to the setting of SRAMWTSC.WTEN. If SRAMWTSC.WTEN = 1, the read will be delayed by one cycle.

57.3.9 Access to the ECC Region

When ECC function is disabled, ECC region can be used as data region.

When ECC function is disabled, direct access is possible. When ECC function is enabled, direct access is disabled. When bypass is enabled, direct access to ECC bit is possible.

Bypass	ECC	Access region	SRAM region	ECC region	Note
enable	—	to SRAM region	—	✔	—
—	—	to ECC region	—	—	Access invalid
disable	enable	to SRAM region	✔	✔	—
—	—	to ECC region	—	—	Access invalid
—	disable	to SRAM region	✔	—	—
—	—	to ECC region	—	✔	—

Note: When using bypass, set SRAMCRn.ECCMOD[1:0] = 00b. See section 57.2.7. SRAMCRn : SRAM Control Register n For ECC RAM (n = 0 to 3).

57.4 Usage Note

57.4.1 Instruction Fetch from SRAM area

When using SRAM to operate a program, initialize the SRAM area so that the CPU can correctly prefetch data. ECC error might occur if the CPU prefetches from an area that is not initialized. Initialize the additional 12-byte area from the end address of the program with the 8-byte boundary. Renesas recommends using the NOP instruction for data initialization.

57.4.2 Notes on Using Error Checking of SRAM

Data in SRAM are undefined when the power is turned on. Therefore, ECC errors occur if the data are read before initialization. The SRAM are read in 8-byte (64-bit) units. Initialize them on 8-byte boundaries.